

X-ray Constrained (Restrained) Wavefunctions

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1. Introduction

This lecture is about wavefunctions derived from experiment, specifically electronic wavefunctions which depend on the positions of the atomic nuclei; and which are fitted to “structure factors” obtained from X-ray diffraction experiments. It could also be seen as possibly the most ludicrously complicated way ever conceived by the human mind to model the X-ray diffraction experiment.

I call this field *ab initio* quantum crystallography by analogy with *ab initio* quantum chemistry, and that field was reviewed by Grobowsky et al.^{1,2} In quantum chemistry the word *ab initio* means “from the beginning”, and there it indicates that basic quantum principles are used to calculate the wavefunction; The word *ab initio* isn’t strictly applicable here since we are combining experimental data into an *ab initio* wavefunction (!) but these days, quantum crystallography is a broad field.^{3,4}

Let’s take a step back: is it even sensible to obtaining a wavefunction from experiment?

Some would say not. They would say the wavefunction is not directly measurable. They would say the wavefunction is used only to make predictions. This is correct. Nevertheless, a model wavefunction can still be *reconstructed* from, or *fitted* to, experimental data. We have done it, and we are doing it.⁵⁻⁸

Besides, the idea to obtain wavefunctions from experiments is very old,⁹ and is currently a very active field in fundamental physics.^{10,11}

However, I need to mention a very important caveat: the wavefunction is not appropriate modelling ensembles, or statistical systems, in which many quantum states are “populated” with a certain probability.¹² The notion of experimental wavefunctions only applies to parts of a system which can be isolated and kept at a relatively low temperature. For us, it means that we are talking about *electronic* wavefunctions at around room temperature, or below. Crystals and even low temperature gases are fine.

A second important question: why would you want to obtain an experimental wavefunction? What can you do with it?

Obtaining accurate wavefunctions from calculations is very time consuming. Apart from obtaining relevant results *more quickly* on the system being studied, the hope is that an experimental wavefunction would be useful for making predictions *for properties not directly related to those being measured*. After all, the wavefunction is supposed to embody all the information in a system; it can be used to predict any property.

This all sounds amazing! So has anything useful been done with experimental wavefunctions, especially those obtained from X-ray experiments?

Probably the major achievement of the wavefunction fitting methods has just been described by S. Grabowsky in the previous lecture: to obtain accurate hydrogen (H) atom positions from X-ray diffraction in statistical agreement with neutron diffraction data.¹³ This is very

exciting, since H atoms occupy the surface of molecules and are critical to chemistry and biology: think about pH, and how important that is in chemistry, for example.

However, apart from this, I do not see any more great applications made with experimental X-ray wavefunctions. Why not?

To answer this question I would ask the reader: are you interested in experiments which are unrelated to the ones you do, or the ones you model? Unrelated to diffraction? Are you interested in NMR experiments? Optical spectroscopy? ESR experiments? Polarised neutron diffraction? Do you do these experiments; are you very familiar with them?

If you think these questions are strange and hard, then the explanation lies here: once you have obtained a fitted wavefunction from one experiment, for it to be truly useful, you must then demonstrate that it can make predictions for experiments unrelated to the one you used to fit the wavefunction. That means you have to work in more than one field—which is more work than normal. And then you must convince your colleagues in both fields that what you are doing might be useful. Even more work.

But for sure, it is very interesting.

The wavefunction is like the Philosopher's Stone: elusive and difficult to master.

So in this chapter I am mainly going to talk about fitting the electronic parts of the electronic wavefunction to X-ray data, explaining how to solve the problem of having too many parameters. I will give some basic examples of things that might be done with the method in the talk, but not in these notes. But if this method is to be developed further, the rest is up to you.

In the laboratory tomorrow, for those who are interested, I will explain how to change the program to do new things. Of course, we will try the method out on existing data, or your own, if you have any. It's fun to see the pictures.

However, here in these notes, I want to begin with the basics. And I want to spend a bit of time on these basics, even if not in the accompanying lecture, because I think these aspects of quantum theory are poorly understood; and because I think it will help with your intuition. You will have to be familiar with multidimensional calculus, and linear algebra, with basic notions of quantum mechanics and classical mechanics.

My approach is historical; I have found the original papers, although confusing, still very rewarding to see and read, since the early guys (footnote)¹ also did not know what was going on—they were just discovering it. But, it will not be a disaster if you skip to the fourth section describing the experimental wavefunction reconstruction technique, and come back to the other stuff later.

¹This is intended to be a genderless descriptor

2. Basics: What is a wavefunction?

Quantum mechanics is one of the fundamental theories of physics. As a matter of fact, the publication in *Ann. de Physique* by de Broglie¹⁴ outlining his idea of matter waves, which was the birth of quantum mechanics, is this year one hundred years old, which is rather special. (Strictly speaking, de Broglie's thesis was from 1924, see Fig. 1, but let's not be pedantic).

The *wavefunction* describes the state of a system in quantum mechanics. The wavefunction holds all the information needed to describe it in detail.

Thus you might say the information in the system includes the locations of all the particles, except that since particles are regarded as waves, the notion of locations doesn't make sense.

By contrast, in Newton's mechanics the object which describes all the system information is its *trajectory curve in phase space*, phase space being composed of all the particle coordinates and velocities (or momenta), depending on whether you use the Lagrangian (or Hamiltonian) formalism.

The wavefunction is usually given the Greek symbol

$$\Psi$$

I suppose this because part of this symbol looks like a wave. This wavefunction is in general complex-number valued, and it is a function of the particle coordinates in the system,

$$\Psi \equiv \Psi(\mathbf{x}_1, \dots, \mathbf{x}_{N_e}; \mathbf{R}_1, \dots, \mathbf{R}_{N_N}; t) \equiv \Psi(\mathbf{x}; \mathbf{R}; t) \quad (1)$$

We are here interested in chemical systems so the $\mathbf{x}$'s are electron coordinates, of which there are N_e . The $\mathbf{R}$'s are nuclear coordinates, which there are N_N . These nuclei are normally assumed to be classical point-like particles in the first instance, which is called the *Born-Oppenheimer approximation*. However, we will later see that in X-ray crystallography this assumption is untenable. The number of electrons and the number and type of nuclei must be decided beforehand based on the molecule or unit cell we are interested in modelling. This is called the *system approximation* which is always made, but rarely mentioned. Finally, Ψ also depends on time t because, well, the wavefunction and the particles it describes can move around, in general. However, we shall generally ignore time here since the systems we are interested in will be stable and time-independent.

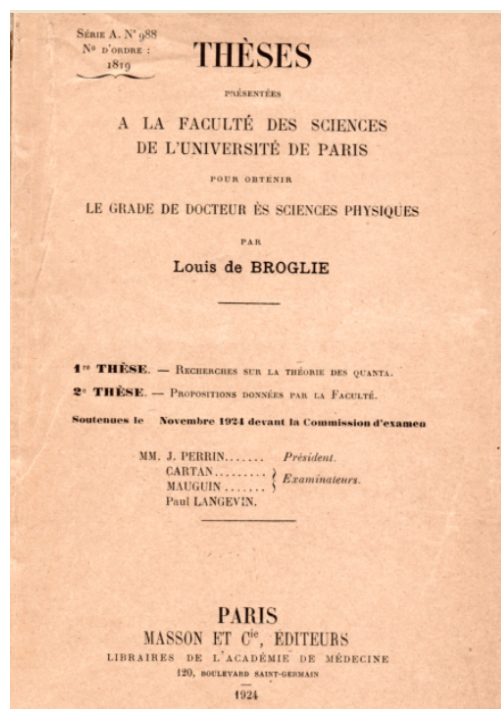


Figure 1: de Broglie's PhD thesis.

Questions and confusions about the wavefunction

There are a few things to be said about the wavefunction which you may already know; many of these are among the most confusing things about quantum mechanics. If you remain confused, never mind, you are in good company: the physicist Richard Feynman said that trying to interpret quantum mechanics would “get you into a blind alley from which nobody has yet escaped”.¹⁵ Still, we must try.

Wavefunctions, probabilities and densities

The most common interpretation of the wavefunction, according to Born,¹⁶ is that the square of the wavefunction tells the probability of finding the particles at their given coordinates. Since probabilities add up to one, this implies that

$$\int |\Psi|^2 d\mathbf{x}_1 \dots d\mathbf{x}_{N_e} = 1. \quad (2)$$

Interestingly, this “Born interpretation” only appears in a footnote of his paper,¹⁶ he actually had the wrong interpretation in the body of the text! (footnote)²

If we are talking about electrons, which are all identical, we may stop the above integration for all but one electron, and define an electron probability density,

$$\begin{aligned} p_e(\mathbf{x}_1; \mathbf{R}) &= \int |\Psi(\mathbf{x}; \mathbf{R})|^2 d\mathbf{x}_2 \dots d\mathbf{x}_{N_e}, \\ \int p_e(\mathbf{x}_1; \mathbf{R}) d\mathbf{x}_1 &= 1. \end{aligned} \quad (3)$$

Since all electrons are indistinguishable from each other, we might as well multiply this single-electron probability by the number of electrons to get the density of all electron in the system,

$$\begin{aligned} \rho_e(\mathbf{x}; \mathbf{R}) &= N_e p_e(\mathbf{x}; \mathbf{R}), \\ \int \rho_e(\mathbf{x}; \mathbf{R}) d\mathbf{x} &= N_e. \end{aligned} \quad (4)$$

The *electron density* ρ_e plays a crucial role in X-ray diffraction, because the strength of X-ray scattering from a point in space is, in the first approximation, proportional to the electron density at that point in space.¹⁸

Note that the electron density depends also on the nuclear coordinates $\mathbf{R}$ of the atoms.

Other particles scatter from different property densities. For example, electron densities

²Schrödinger actually has nearly the same interpretation in his original paper,¹⁷ eqn. (27), which he calls a “fluctuating charge”.

scatter from the electric potential density, neutrons scatter from the density of nuclear electron density, and polarised electrons and neutrons scatter, in addition to other things, also to any local magnetic field densities in the material. Actually, even circularly polarised X-rays also scatter from the local magnetic fields.

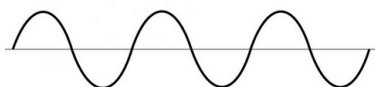
How can electrons have coordinates when they are waves?

The coordinates of any particle in the wavefunction are not “positions” — even though they are often talked about in that way — especially the nuclear coordinates, which in the Born-Oppenheimer approximation actually are nuclear positions! Confusing!

Anyway,

Wavefunction coordinates are not positions but simply dummy variables which are used to describe the shape of the wavefunction.

To illustrate this idea consider the simple wave shape below.



The coordinate corresponds to the grey axis, but it is only used as a means to describe the shape of the wave, which is described by a sine or cosine function.

Moreover, if you want to know the position of a particle, you must perform an experiment, and the resulting measured position for systems prepared in an identical state Ψ will not always be the same: this is the famous uncertainly principle, which is intimately linked to the fact that particles are described by distributed wavelike functions.

More than three coordinates?

Even more confusing, is that coordinates are not just composed of three numbers $\mathbf{r}$, but have a fourth component called the spin coordinate.¹⁹ Thus a typical electron coordinate would be written

$$\mathbf{x} = (\mathbf{r}, \xi). \quad (5)$$

where $\mathbf{r}$ is the spatial coordinate, and the last number ξ is the spin coordinate. Whereas $\mathbf{r}$ describes how the wavefunction looks in “normal” three-dimensional space,

The spin coordinate describes the shape of a particle wavefunction when it is rotated, in its own private dimension.

While it is true that we can already rotate objects in normal three dimensional space, in quantum mechanics there is an additional and separate coordinate ξ attached to each particle

which describes how it rotates *separately to any rotation in normal space*. Wild, but true! (footnote)³ (footnote)⁴ (footnote)⁵.

Nuclei also have a spin coordinate, just like electrons, but here we shall treat nuclei (mostly) as classical particles and ignore any spin coordinate. This approximation works well, unless you are looking for such effects (like in NMR experiments) of possibly for H atoms, the lightest nuclei

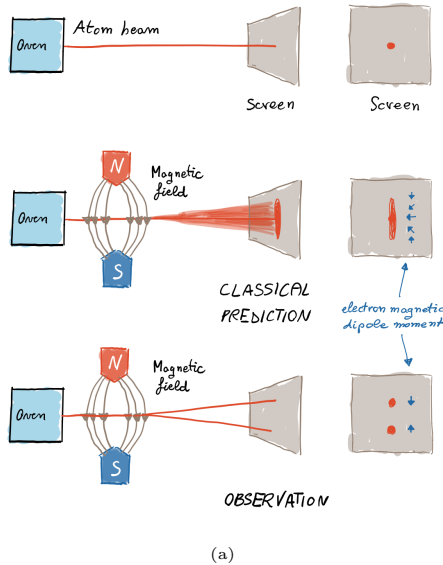
“Zur Quantenmechanik des magnetischen Elektrons,” Zeit. f. Phys. **43** (1927), 601-623.

On the quantum mechanics of magnetic electrons

By W. PAULI, Jr. in Hamburg

(Received on 3 May 1927)

Translated by D. H. Delphenich



moment in a fixed direction as further independent variables in order to carry out the computations of its rotational degrees of freedom, along with the position coordinates of any electron. In contradiction to classical mechanics, these variables can assume only the variables $+\frac{1}{2}\frac{h}{2\pi}$ and $-\frac{1}{2}\frac{h}{2\pi}$, which is completely independent of any sort of external field. The appearance of the aforementioned new variables thus implies a simple splitting of the eigenfunctions into two position functions ψ_α , ψ_β for one electron, and more generally, for N electrons they split into 2^N functions, which are to be regarded as the “probability amplitudes” that in a well-defined stationary state of the system not only do the position coordinates of the electrons lie in a given infinitesimal interval, but also that the components of their proper moments in the chosen direction should have the given values, which are $+\frac{1}{2}\frac{h}{2\pi}$ for ψ_α and $-\frac{1}{2}\frac{h}{2\pi}$ for ψ_β . Methods

$$\left. \begin{aligned} \psi^{\text{sym}} \alpha_1 \alpha_2(q_1, q_2) &= \psi^{\text{sym}} \alpha_1 \alpha_2(q_2, q_1), \quad \psi^{\text{sym}} \alpha_1 \beta_2(q_1, q_2) = \psi^{\text{sym}} \beta_1 \alpha_2(q_2, q_1), \\ \psi^{\text{sym}} \beta_1 \alpha_2(q_1, q_2) &= \psi^{\text{sym}} \beta_1 \alpha_2(q_2, q_1), \quad \psi^{\text{sym}} \beta_1 \beta_2(q_1, q_2) = \psi^{\text{sym}} \beta_1 \beta_2(q_2, q_1), \end{aligned} \right\} \quad (26')$$

$$\left. \begin{aligned} \psi^{\text{antis}} \alpha_1 \alpha_2(q_1, q_2) &= -\psi^{\text{antis}} \alpha_1 \alpha_2(q_2, q_1), \quad \psi^{\text{antis}} \alpha_1 \beta_2(q_1, q_2) = -\psi^{\text{antis}} \beta_1 \alpha_2(q_2, q_1), \\ \psi^{\text{antis}} \beta_1 \alpha_2(q_1, q_2) &= -\psi^{\text{antis}} \beta_1 \alpha_2(q_2, q_1), \quad \psi^{\text{antis}} \beta_1 \beta_2(q_1, q_2) = -\psi^{\text{antis}} \beta_1 \beta_2(q_2, q_1), \end{aligned} \right\} \quad (27')$$

The skew-symmetric solution is also the one that fulfills the “equivalence rule” in the general case of N electrons, and is the only one that occurs in nature². It seems to me to

Figure 2: (a) The Stern & Gerlach 1922 experiment²¹ gave the first indication that electrons might be magnetic, and that this magnetism was quantised in only two directions. (b) Key extracts from a translation of Pauli’s 1927 paper²² which describe “up” (α) and “down” (β) spins. There is no good reason I can see why they are called α and β ! Underlined is the Nobel prize winning line, where the antisymmetry requirement is stated for the electronic wavefunction. Antisymmetry explains why electrons “fill up” orbitals two at a time; which then explains the whole periodic table. Note that Dirac²³ had already worked out in 1926 that: “the symmetrical eigenfunctions alone or the antisymmetrical eigenfunctions alone give a complete solution of the problem. The theory at present is incapable of deciding which solution is the correct one”. In fact, theory cannot determine this fact, only experimental induction. And Pauli stated it. See also footnote 6.

³You may have heard of “string theory” in physics? In that theory, it is posited that there are additional translational coordinates attached specifically to each particle, in addition to $\mathbf{r}$ (and just like the spin coordinate ξ) but there is no experimental evidence for these yet.

⁴Also, I should say, that the “spin coordinate” is actually three coordinates, because to describe a rotation, you need an axis of rotation (which requires two parameters) as well as a parameter to describe how much rotation to apply (which is another parameter). But this is too much information in most cases since in the end all that is required for calculations is known how to integrate over the spin coordinate, the fact that there are two possible discrete values, and how the spin coordinate(s) behave under rotation; the last two things are closely connected.

⁵The idea that approximate symmetries of a system (such as translation or rotation) may be used to define coordinates for that system has its origin in classical mechanics, via the work of E. Noether; but its full power was developed largely by S. Coleman, who delivered many lectures on this topic here in Erice, in the School of Physics organised by A. Zichichi. Coleman’s lecture notes “Aspects of Symmetry”²⁰ are incomprehensible to me except for the foreword, which is quite funny, and worth reading.

Spin takes on discrete values when measured

All particles have a unique spin coordinate; but unlike translation, this spin is not obviously manifested in the macroscopic life we are familiar with. Moreover, unlike position, which when measured may have any continuous value,

The spin coordinate when measured may take only distinct values, typically called “up” or α , or “down” or β .

In particular, the electron spin coordinate, when “measured” via its associated magnetic moment, yields only two possible values. Nevertheless, the spin coordinate ξ is a continuous dummy variable just like the position coordinate $\mathbf{r}$, and we can integrate over it, as was already assumed in equation (2).

The classic experiment which suggests the occurrence of the spin coordinate is the Stern-Gerlach experiment, Fig. 2a.

The original paper by Pauli where he derived the properties of the α and β values of the spin coordinate is shown in Fig. 2b.

Incidentally, the reason the spin coordinate is has the confusing name “spin” is that, originally, it was thought that electrons were actually spinning (footnote)⁶

How do you find out what the wavefunction looks like?

Schrödinger derived an equation which describes the time-dependent *dynamical behaviour* of the wavefunction. See Fig. 3 for extracts from his paper.

In modern notation, the time-independent Schrödinger equation in the Born-Oppenheimer approximation with fixed nuclear positions is

$$H\Psi = E\Psi. \tag{6}$$

Here E is an allowed energy of the system, and H is the Hamiltonian energy operator for

⁶The idea that electrons are spinning goes back at least to Compton²⁴ (yes, the X-ray guy), before quantum mechanics. Goudsmit and Uhlenbeck were the ones awarded the Nobel prize for “discovering” electron spin. What they really discovered was that a “spin quantum number” should be ascribed to a rotation of the electron, in order to explain many spectroscopic results²⁵. Even though Pauli²² used rotations everywhere to describe the electron, see Fig. 2b, he did not firmly connect his result to experiment; actually he relied very much on the work of Stoner,²⁶ who did not receive any prize! But, Pauli was awarded the Nobel prize not only for the permutation symmetry of the wavefunctions, but more importantly also for a proof that the permutation symmetry in the wavefunctions of identical particles depends on their spin quantum number—the truly remarkable spin-statistics theorem. For more on this, with translations and commentary on the original papers, see the book by Duck & Sudarshan.²⁷

(q -space, not pq -space). We shall effect this generalization in a somewhat different manner in Section 7. Using the usual notations the kinetic energy T is

$$T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) = (1/2m)(p_x^2 + p_y^2 + p_z^2). \quad (1)$$

The well-known Hamiltonian function of action W ,

$$W = \int_{t_0}^t (T - V) dt, \quad \text{This is the principle of least action} \quad (2)$$

taken as a function of the upper limit t and of the final values of the coordinates x, y, z satisfies the Hamiltonian partial differential equation,

$$\partial W / \partial t + (1/2m)[(\partial W / \partial x)^2 + (\partial W / \partial y)^2 + (\partial W / \partial z)^2] + V(x, y, z) = 0. \quad (3)$$

To solve this equation, we put as usual

$$W = -Et + S(x, y, z), \quad \text{This is the beginning of the "separation of variables" method and S is the "reduced action"} \quad (4)$$

E being an integration constant, viz., the total energy, and S a function of x, y, z only. Eq. (3) may then be written

$$|\text{grad} W| = [2m(E - V)]^{1/2}. \quad \text{Here Schrödinger notes that W forms "wavefronts" which move with velocity u, which was also noticed earlier by Hamilton} \quad (5)$$

In this form it lends itself to a very simple geometrical interpretation.

Now let the time vary, Eq. (4) shows that the system of surfaces will not vary, but that the values of the constants will travel along the normals from surface to surface with a certain speed u , given by

$$u = E / [2m(E - V)]^{1/2}. \quad (6)$$

The velocity u is a function of the energy-constant E and besides, since it contains $V(x, y, z)$ is a function of the coordinates.

(a)

3. Let us return to the system of W -surfaces, dealt with in Section 1 and let us associate with them the idea of stationary sinusoidal waves whose phase is given by the quantity W , Eq. (4). The wave-function, say ψ , will be of the form

$$\begin{aligned} \psi &= A(x, y, z) \sin(W/K) \\ &= A(x, y, z) \sin[-Et/K + S(x, y, z)/K], \end{aligned} \quad (9)$$

A being an "amplitude" function. The constant K must be introduced and must have the physical dimension of *action* (energy $\times$ time), since the argument of a sine must always be a pure number. Now, since the fre-

As stated above, the wave-phenomena must in this case be studied in detail. This can only be done by using an "equation of wave propagation." Which one is this to be? In the case of a single material point, moving in an external field of force, the simplest way is to try to use the ordinary wave-equation

$$\Delta\psi - \ddot{\psi}/u^2 = 0 \quad (15)$$

and to insert for u the quantity given by Eq. (6), which depends on the space coordinates (through the potential energy V) and on the frequency E/h . The latter dependence restricts the use of (15) to such functions ψ as depend on the time only through the factor $e^{\pm 2\pi i E t/h}$. (A similar restriction is always imposed on the wave equation, as soon as we have dispersion.) So we shall have

$$\ddot{\psi} = -4\pi^2 E^2 \psi / h^2$$

Inserting this and Eq. (6) in Eq. (15) we get

$$\Delta\psi + 8\pi^2 m(E - V)\psi / h^2 = 0, \quad (16)$$

where ψ may be assumed to depend on x, y, z only. (We omit changing the notation of the dependent variable, which we really ought to do.)

(b)

Figure 3: Key extracts from Schrödinger's 1926 paper "An Undulatory Theory of the Mechanics of Atoms and Molecules" ¹⁷ (a) The classical mechanics "principle of least action" noting Hamilton's optico-mechanical analogy, where wave fronts in W move with velocity u . (b) In eqn. (9) Schrödinger writes $\Psi = \Psi_0 e^{iS'/\hbar} = \Psi_0 e^{S/\hbar}$ [he does this later in the paper, before eqn (27)] and substitutes that into the classic wave eqn. (15) with a space-dependent velocity u , resulting in the final eqn. (16). Note that the reduced action S was allowed to become complex; the wavefunction is related to the logarithm of S . Later in this paper Schrödinger also derives his equation by a variation principle; but this first way has more insight. For background in classical mechanics I recommend the book by Goldstein et al ²⁸ which is hard to beat. For a modern derivation of the Schrödinger equation from symmetry ideas, see Ballentine's book ¹⁹

that same system of electrons and nuclei,

$$H = \frac{1}{2} \sum_{i=1}^{N_e} \nabla_i^2 - \sum_{i=1}^{N_e} \sum_{A=1}^{N_N} \frac{Z_A}{|\mathbf{r}_i - \mathbf{R}_A|} + \sum_{i=1}^{N_e} \sum_{j < i}^{N_e} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} \quad (7)$$

This equation may have multiple solutions, and the allowed energies may be both discrete and continuous.

The Schrödinger is a partial differential eigenvalue equation in the electron coordinates. In contrast simple equations whose solutions are numbers, the solution to the Schrödinger is a whole function Ψ .

Schrödinger's equation is analogous to Newton's equation in classical mechanics; indeed, Schrödinger derived his equation from classical mechanics in the Hamilton-Jacobi form, by allowing the action to become a complex number quantity.

While Newton's laws tell us how to calculate the dynamical motion of particles in phase space when they are acted on by forces (such as electrical or magnetic forces, which are the

most important ones in chemistry) the Schrödinger equation tells us how to calculate the dynamical wiggling of the wavefunction subject to the same forces (footnote)⁷.

Below I describe a different way to reconstruction a wavefunction using the direct input of experimental information.

Exchange symmetry of wavefunctions for identical particles

Schrödinger's equation is not completely correct. For a start, it does not properly take into account the spin coordinate. Second, it does not work for systems with more than one identical particle—which if you think about it, is almost everything!

Dirac solved both these problems, and in addition, the problem of combining relativity theory with quantum mechanics (to a certain extent). He shared the Nobel prize with Schrödinger (footnote)⁸. Since relativistic effects are energetically relatively small, we will stick with the Schrödinger's approach.

The problem is (as noted by Dirac) wavefunctions involving two or more identical particles must be either symmetric or antisymmetric when the coordinates of those identical particles are exchanged. Thus, for a two-identical-particle wavefunction, we must have

$$\Psi(\mathbf{y}_1, \mathbf{y}_2) = \pm \Psi(\mathbf{y}_2, \mathbf{y}_1), \quad (8)$$

The reason often given for the above relationship is that, according to Born, the square of the wavefunction $|\Psi(\mathbf{y}_1, \mathbf{y}_2)|^2$ represents the probability of finding an electron at $\mathbf{y}_1$ and another at $\mathbf{y}_2$; however, since the particles are identical, we must have the same probability when the coordinates of the two particles are swapped i.e. we must have $|\Psi(\mathbf{y}_1, \mathbf{y}_2)|^2 = |\Psi(\mathbf{y}_2, \mathbf{y}_1)|^2$. Taking the square root of this equation gives the above result. I do not think this is a good argument; Dirac gave a better argument in terms of transitions between electronic states.

It turns out that for electrons, the wavefunction must be antisymmetric,

$$\Psi(\mathbf{x}_1, \mathbf{x}_2) = -\Psi(\mathbf{x}_2, \mathbf{x}_1). \quad (9)$$

This cannot be deduced from theory, it must be induced from experiments.

The most important experimental evidence for this hypothesis is that, the whole periodic table is explained by it, since in most cases the antisymmetry condition above only allows two electrons to “occupy” the same spatial region, provided they have opposite spin. So electron spin really does behave like an extra dimension—but a discrete dimension, additional to the usual three we are used to. Chemistry depends on it.

⁷Schrödinger discovered his equation only one year after de Broglie's work was published; and he made his discovery while on holiday with a girlfriend. But certainly, Schrödinger and many of his contemporaries did not waste any time!

⁸It is amazing to think that Dirac wandered around here, in Erice. His museum is nearby.

Anyway, the above equation is known as the Pauli exclusion principle, since he first enunciated it for electrons, and noticed that it was connected with the electron spin coordinate, as we mentioned; see Fig. 2b.

3. Quantum chemistry: calculating wavefunctions

Before we learn about fitting wavefunctions to experimental data, we need to know how to calculate them from the Schrödinger equation. It will be very brief but I still hope it will be enough for you to get more than just a flavour of what is going on.

The Schrödinger equation is very complicated; it isn't possible to solve it exactly except for systems with two or three particles (footnote)⁹. Therefore, if we are to progress for larger systems, approximations must be made. Luckily it is possible to systematically remove these approximations to obtain, more accurate results, albeit with increasingly impractical cost.

The key approximations are: the spinorbital product wavefunction; expansion of the spinorbitals in a basis set of fitting functions; and the use of the variational theorem.

The Hartree wavefunction: the simplest possible

The oldest approximation, and still one of the most important, is the Hartree product wavefunction

$$\Psi \approx \Phi = \phi(\mathbf{x}_1) \dots \phi(\mathbf{x}_{N_e}) \quad (10)$$

The Hartree wavefunction comprises just a product of *spinorbitals* ϕ_i , each spinorbital being a function of a just a single electron's coordinates (including the spin coordinate). Hence, spinorbitals are also called *one-electron spinorbitals*. In molecules they are called *molecular spinorbitals*. To ensure that the wavefunction is normalised, equation (2), it is usually assumed that the spinorbitals are orthonormal

$$\int \phi_i(\mathbf{x}) \phi_j(\mathbf{x}) d\mathbf{x} = \delta_{ij}. \quad (11)$$

Here δ_{ij} is the Kronecker delta, equal to one if i and j are the same, or zero otherwise.

The spinorbitals are not *required* to be orthonormal; and in unrestricted theories which are useful for magnetic systems, the spatial parts of these spinorbitals may not be orthonormal, as we discuss below. Since orthonormality of the spinorbitals is not mandatory, the above equation must be enforced via a constraint, which leads to the appearance of the orbital energies, which are in fact Lagrange multipliers associated with enforcing the above rule.

⁹Contrary to urban myth, the three-body problem is not insoluble; see the work of Maslen and coworkers.²⁹ There has even been a recent breakthrough using an integral equation approach from scattering theory.³⁰

As a matter of fact, when Schrödinger derived his equation, in the second section of his paper, he used a constraint associated with the normalisation condition (2), and that Lagrange multiplier turned out to be the energy of the system E in his wave equation.

Anyway, Hartree solved for his orbitals in eqn. (12) in 1928, via an iterative self-consistent field procedure (SCF) which is still used today.^{31,32}

He used mechanical computing machines, and later, the first electronic computing machines.

Hartree had been already interested in X-ray scattering since at least 1925.³³

So it was not surprising that in that same year that he described his SCF method James, Waller & Hartree applied his SCF method to the NaCl crystal to show that zero-point motion of atoms could be detected in the nuclei of that crystal, see Fig. 4.

Waller is the same one, from the Debye-Waller factor in X-ray crystallography.

This shows the close connection in the early years between quantum mechanics and X-ray diffraction, something I stressed in my lecture at the inaugural conference on QCr at Erice, and likewise, eloquently by Macchi.³⁴ Of course, there was also a close connection between these topics and computing machines. (footnote)¹⁰

The Hartree-Fock wavefunction: the next simplest

The problem is, Hartree's product wavefunction does not obey the Pauli principle. It had only just been discovered!

Luckily, it is easy to satisfy Pauli's antisymmetry principle. All that is required is to sum together Hartree product wavefunctions which differ only in the fact that the electron coordinates in each are permuted, multiplying each permuted term by a positive sign for an "even" permutation, or a negative sign for "odd" permutation. The result is the single determinant wavefunction, written

$$\Psi \approx \Phi = \mathcal{A}[\phi(\mathbf{x}_1) \dots \phi(\mathbf{x}_{N_e})] \quad (12)$$

¹⁰Hartree was one of the first to use electronic computing machines, and of course he came across Alan Turing. It is fascinating to see that these two argued over many years about whether computers could be conscious, a topic of great interest today; and their argument led to the famous "Turing test".³⁵

An Investigation into the Existence of Zero-Point Energy in the Rock-Salt Lattice by an X-Ray Diffraction Method.

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1. Introduction.

In the present paper we shall attempt to collate the results of four separate lines of research which, taken together, appear to provide some interesting checks between theory and experiment. The investigations to be considered are (1) the discussion by Waller* and by Wentzel,† on the basis of the quantum (wave) mechanics, of the scattering of radiation by an atom; (2) the calculation by Hartree of the Schrödinger distribution of charge in the atoms of chlorine and sodium; (3) the measurements of James and Miss Firth‡ of the scattering power of the sodium and chlorine atoms in the rock-salt crystal for X-rays at a series of temperatures extending as low as the temperature of liquid air: and

Figure 4: Could this be the first paper in quantum crystallography? And who is Miss Firth?!

The $\mathcal{A}$ symbol is the *antisymmetriser* which performs the just described permutations-with-sign-change operation. Dirac noted that this wavefunction could be written in the form of a determinant,²³ which explains why it is also called a determinant wavefunction. Though for some reason people more commonly call the above a Slater determinant. Since there are $N_e!$ terms in this wavefunction, the $\mathcal{A}$ symbol also includes a normalisation factor of $(N_e!)^{-1/2}$.

To make the antisymmetrisation more clear, consider the example of a three electron determinant wavefunction: it can be written explicitly as

$$\Phi = \frac{1}{\sqrt{3!}} [\phi_1(\mathbf{x}_1)\phi_2(\mathbf{x}_2)\phi_3(\mathbf{x}_3) - \phi_1(\mathbf{x}_1)\phi_2(\mathbf{x}_3)\phi_3(\mathbf{x}_2) - \phi_1(\mathbf{x}_2)\phi_2(\mathbf{x}_1)\phi_3(\mathbf{x}_3) + \phi_1(\mathbf{x}_2)\phi_2(\mathbf{x}_3)\phi_3(\mathbf{x}_1) - \phi_1(\mathbf{x}_3)\phi_2(\mathbf{x}_2)\phi_3(\mathbf{x}_1) + \phi_1(\mathbf{x}_3)\phi_2(\mathbf{x}_1)\phi_3(\mathbf{x}_2)] \quad (13)$$

You can see why we do not want to write down such wavefunctions explicitly (footnote)¹¹.

Closed-shell wavefunctions and molecular orbitals

The most common systems in nature are closed-shell systems, composed of an even number of electrons; and where all electrons are paired up (footnote)¹².

In the paired-up even-electron closed-shell case a further approximation, called the restricted spinorbital approximation, is used for the the spinorbitals—namely that they come in pairs composed of the product of a common spatial part, and either a spin part which is either “up” or “down”,

$$\phi_I(\mathbf{x}) = \begin{cases} \varphi_i(\mathbf{r})\alpha(\xi), & \text{or} \\ \varphi_i(\mathbf{r})\beta(\xi). \end{cases} \quad (14)$$

The functions $\alpha(\xi)$ and $\beta(\xi)$ are defined by symmetry; we do not write down their form, but note that they form an orthonormal pair; and as a consequence of equation (11) the $\{\varphi_i(\mathbf{r})\}$ form an orthonormal set of $N_e/2$ functions called *spatial orbitals*, *orbitals*, or *molecular orbitals* (MOs).

For odd-electron systems, we can allow the spatial parts of the spinorbitals to be differ-

¹¹Even though Dirac wrote down the determinant form of the wavefunction first, it was Fock (1930) who first published results including the effect of antisymmetry in determinant wavefunction calculations before Hartree could (according to Froese Fischer³⁶) so if the spinorbitals are determined using the Hartree’s SCF method (explained below) they are called Hartree-Fock spinorbitals

¹²The reason for this pairing up has nothing to do with the magnetic forces between the spins; it has instead to do with the fact that two electrons with opposite spin can occupy the same spatial region, especially any region close to large positively charged nucleus, which is electrically rather attractive, at the expense of putting up with the repulsion of the other electron; which being light, can arrange itself to be “on the other side” of the nucleus if needed. This is the reasoning needed to explain the electronic structure of atoms, and hence the periodic table. As a matter of fact this same approach explains why odd-electron molecules are rare in nature, and also rather reactive: they prefer to form even electron molecules.

ent; and in this case we would have the *unrestricted spinorbital approximation*. Such an approximation is useful for systems with unpaired spins, which are magnetic.

It is also possible to write the spinorbitals to be a linear combination of the α and β parts, which would be the *general spinorbital approximation*.³⁷ Such general spinorbitals are called *two-component spinors*.

Here, we are only be concerned with restricted spinorbitals.

Molecular orbital coefficients: basis functions

We now come to a very important and practical way to describe the spatial orbital part of the restricted spinorbitals.

The idea is to fit the molecular with a set of N_{bf} *basis functions*. Then to change the shape of the molecular orbital, all we have to do is use different fitting coefficients. This is called the *linear combination of basis functions method*, written as

$$\varphi_i(\mathbf{r}) = \sum_{\mu=1}^{N_{\text{bf}}} G_{\mu}(\mathbf{r}) c_{\mu i} \quad (15)$$

where $G_{\mu}(\mathbf{r})$ are the one-electron fitting functions, and where the $\mathbf{c}$ is the matrix of *molecular orbital coefficients* of size $N_{\text{bf}} \times (N_{\text{e}}/2)$.

Due to the orthonormality requirement on the molecular orbitals, not all the MO coefficients are independent. The number of independent parameters needed to describe the MOs is given by N_{param} of equation (19).

The molecular orbital coefficients may be determined using the *variational principle* discussed below. Essentially, this principle allows to determine the coefficients by minimising an energy function.

The basis functions are sometimes called atomic orbitals but this old fashioned terminology is very confusing and should not be used today.

Gaussian basis functions and basis sets

Basis functions may be any functions you like—they are simply functions that will be used to “fit” the molecular orbitals. The more they look like the exact Hartree-Fock orbitals, the better they will be.

In most cases, however, for molecules, and for practical reasons, the basis functions are taken

to be atom-centred three dimensional Gaussian functions multiplied by a polynomial factor,

$$G(\mathbf{r}; \alpha; \mathbf{l}; \mathbf{R}) = (r_1 - R_1)^{l_1} (r_2 - R_2)^{l_2} (r_3 - R_3)^{l_3} e^{-\alpha|\mathbf{r}-\mathbf{R}|^2}. \quad (16)$$

Here

$\mathbf{R}$ is the “center” of the Gaussian basis function, which is usually placed at the position of an atomic nucleus.

α is the Gaussian exponent, or simple the exponent (not to be confuse it with the “up” spin function with the same symbol). The larger the exponent, the “sharper” the gaussian function.

$\mathbf{l}$ are non-negative integers called the cartesian “angular momentum” quantum numbers. They are simply the powers of the cartesian monomial parts of the Gaussian. If all the $\mathbf{l}$ are zero we have a single s-type function; if any one of them is equal to one, we have a p-type functions, of which there are three in total; if the sum of them is equal to two, we have a d-type function, of which there are six in total; if the sum of them is three, we have an f-type function, of which there are ten; and so on.

Different values of the parameters α and $\mathbf{l}$ lead to different basis functions, which are then used as fitting functions.

The cartesian Gaussian functions can be combined to make spherical harmonics, which describe the angular part of any function with maximum efficiency, and nowadays when Gaussian functions are used, it is usually assumed that they are “spherical harmonic Gaussian functions”, unless they are explicitly called “cartesian Gaussian functions”.

A single s-type Gaussian function is a very poor approximation to an atomic s-type wavefunction. This is because s-type wavefunctions have a cusp at the nucleus, and they have an exponential form $e^{-\alpha|\mathbf{r}-\mathbf{R}|}$ not a Gaussian form. However, linear combinations of Gaussian functions (again!) start to form very good approximation to an exponential function, and as was stated, calculations with them are very fast and practical.

Nowadays there are many sets of “canned” basis developed by researchers over the years, that can be used in wavefunction calculations to fit the molecular orbitals. These basis sets which are different for different atoms, and for each atom very specific linear combinations of these “primitive” Gaussian functions of the form above. The expansion coefficients in these linear combinations, along with the exponents, are tabulated, and given very strange names. For example, one of the earliest basis sets called STO-3G, which stands for a “Slater-type exponential orbital expanded with three Gaussians” was derived by Hehre, Stewart and Pople (Bob Stewart is very well known in crystallography since he developed the pseudoatom aspherical model that is very popular in high resolution X-ray diffraction studies).

The most popular Gaussian basis sets are probably the Pople style 6-31G(d), 6-311G(d,p) basis sets, and the Ahlrichs def2-DZVP, def2-TZVP basis sets, which are available for many

atoms in the periodic table. However, there are very few Gaussian basis sets available for all atoms of the periodic table.

Plane-wave basis functions

It is also common to use plane-wave basis functions, especially for calculations on non-molecular periodic crystals. Integrals with these are also extremely easy, but they suffer from the problem that a combination of many of them are required to represent atomic and molecular wavefunctions—especially for electrons near the core of an atom. Since we will be discussing applications to X-ray diffraction where core electrons are very important, we will not consider them further.

Finding the molecular orbitals: variational principle

For many types of wavefunction, and for the Hartree-Fock wavefunction in particular, the method that is used to determine the unknown wavefunction parameters is the *variational principle*³⁷

The variational principle states that the energy of any approximate wavefunction Φ always lies above the energy E of the exact wavefunction Ψ ,

$$E_{\text{QM}}[\Phi] = \int \Phi^* H \Phi d\mathbf{x} \equiv \langle \Phi | H | \Phi \rangle \geq E \quad (17)$$

Moreover, the wavefunction Φ which best approximates the exact wavefunction Ψ is the one which minimises the energy expression $E_{\text{QM}}[\Phi]$.

The energy is a function $E_{\text{QM}}[\Phi]$ is a function of a function, and so we call it a functional, and use square instead of the usual round brackets.

The importance of this result is that we may substitute equation (16), then into equation (15), and that into equation (14), and then that into (12), and finally that into $E_{\text{QM}}[\Phi]$ from equation (17) above; then we can evaluate all the integrals involving Gaussian functions which occur in $E_{\text{QM}}[\Phi]$ (yes, it is possible); and the result will be an expression involving only the MO coefficients $\mathbf{c}$. That is, we can convert $E_{\text{QM}} = E_{\text{QM}}(\mathbf{c})$ into a normal function. Finally, by minimising this energy function, that is by setting its derivatives to zero, we get the celebrated Hartree-Fock equations in a finite basis set! They are shown at the very end.

This is remarkable, because we have converted the extremely complicated Schrödinger partial differential equation into a numerical minimisation problem.

It is not easy, but it is doable.

It becomes much easier if we can program and use computers to organise the calculations.

4. Experimental wavefunctions

There are basically two approaches to obtaining an experimental wavefunction:

1. The reconstructive approach.^{9,38}
2. The weak measurement approach.¹⁰

The weak measurement approach is very recent and involves exactly what the words say: a system that is weakly perturbed in some way is then probed, enabling a measurement of the wavefunction and its phase without destroying it. We will not discuss this technique further, which is more suited to measuring entangled optical states.¹¹

Wavefunction reconstruction methods may also be divided into two types: exact and approximate.

- In exact methods one does not recover the wavefunction itself, but rather the density matrices associated with the state [the density matrix is related to the square of the wavefunction integrated over all but one or two of the particle coordinates, similar to equation (4). The density matrix approach also allows to obtaining information on systems which cannot in principle be described by a wavefunction, such as systems in thermal equilibrium, which involve a Boltzmann distribution over different pure states (a pure state is one that may be described by a wavefunction).
- In contrast to exact density matrix methods, approximate method involve obtaining a *model wavefunction* from the experimental data. These are the methods we are most concerned about here, since they are quite practical. Also, even though these methods are approximate, they can be systematically improved, if the variational method is used to determine the approximate wavefunction.

As stated in the introduction, we are here mainly going to be concerned with electronic wavefunctions for crystalline systems around room temperature or colder where it is appropriate to use wavefunctions, and we are going to use the X-ray diffraction experiment,⁹ for reasons that will be explained in the next section (footnote)¹³.

¹³The wavefunctions for the nuclei in the molecules are not derived, since these are not usually in the ground state: the temperature is too high, they are vibrating, and in multiple quantum states. Nevertheless, it is not enough to treat these nuclei as point particles and simply ignore these vibrations. The nuclei are usually treated as being “smeared out” via a gaussian probability distribution, described by so-called *atomic displacement parameters* (ADPs) which, as you probably know, are refined along with the positions of the atomic nuclei in a typical X-ray diffraction experiment. This smearing out is not only due to the vibrations of the atomic nuclei, but also due to the fact that in different unit cells of the crystal, the atomic nuclei are not quite in translationally equivalent positions—because the crystal is typically composed of thousands or even millions of misaligned crystallites. Indeed, if the crystal were completely perfect—say like in a single crystal of silicon made by zone refining for the computer chip industry—the usual equations for fitting the intensity of X-ray diffraction spots would not apply, since the diffracted X-ray beam would diffract again

Idea to reconstruct a wavefunction from X-ray experiments

Hohenberg and Kohn proved a famous theorem (in only two lines!) that the electron density ρ_e for a non-degenerate ground state uniquely determines the wavefunction.³⁹

So according to quantum mechanics, there is enough information in the electron density to reconstruct a wavefunction. This is rather remarkable since the electron density is only a three dimensional function, and the wavefunction is not!

On the other hand, X-ray experiments scatter from the electron density, and X-ray diffraction experiments provide tens or hundreds of thousands of reflections which are Fourier components of the electron density, so-called *structure factors*.¹⁸

Therefore, the Hohenberg and Kohn theorem, combined with X-ray diffraction experiments, permits us to reconstruct a model wavefunction. All we need to do is obtain a lot of data, make increasingly better wavefunction models, and fit.

Fig. 5 illustrates the the general procedure.

Modelling the X-ray experiment

Here is a single determinant electronic wavefunction in the Born-Oppenheimer approximation:

$$\Phi \equiv \Phi(\mathbf{x}_1, \dots, \mathbf{x}_{N_e}; \mathbf{R}) \quad (18)$$

It is a complicated function which is described by many parameters. Some of the parameters in the wavefunction — the positions of the nuclei $\mathbf{R}$ — can be refined by least squares to experimental data. This is the Hirshfeld atom refinement (HAR) method,^{6,7} and it is assumed to have been performed beforehand.

But what about determining the other parameters in the wavefunction e.g. the MO coeffi-

and again, multiple times. This is multiple diffraction is called *dynamical scattering* (not a very good name in my opinion). Dynamical scattering is very important for electron diffraction because electrons scatter from materials much more strongly than X-rays, due to the fact that electrons are strongly repelled by the other electrons in the material, and as a result they are subject to multiple scattering effects much more than X-rays; though, very intense X-ray reflections are also subject to multiple scattering; we have observed that these are always observed to be smaller than expected based on our calculations; no one has yet taken into account multiple scattering effects in *ab initio* quantum crystallography



Figure 5: Stages in a typical wavefunction reconstruction. (1) Think of a model and an experiment that can determine all the parameters in that model. (2) Carry out the actual experiment and record the observed values. (3) Analyse the data and determine the parameters. This procedure above might be considered a measurement since, in step 2, it often occurs that “model corrected observations” are used, and these corrections are often quite complex. What is to distinguish these corrections from the process above? Indeed, Einstein says: “... on [the] entire long path from a process up to our conscious perception we need to know how Nature is working in order to claim that we have observed anything at all”.⁷

cients in a single determinant wavefunction (if that is the wavefunction to be fitted)?

Levy and Goldstein⁴⁰ noted that a single determinant wavefunction could be obtained from an X-ray experiment provided there are more reflection data than parameters in that wavefunction i.e. a fit could be made only if

$$N_{\text{refl}} > N_{\Phi} = (N_{\text{e}}/2)(N_{\text{bf}} - N_{\text{e}}/2). \quad (19)$$

Here N_{refl} is the number of data and N_{Φ} is the number of parameters needed to describe a single determinant wavefunction. As can be seen, this excess of data is experimentally achievable only for systems with small numbers of electrons; or equivalently for systems where only a small number of basis functions are used; but, more basis functions are needed if there are more electrons!

To actually get the experimental wavefunction one has to just has to maximise the agreement between the experimental data and the corresponding model-calculated values i.e. one has to minimize the square of a “goodness of fit” agreement statistic between calculated and observed data. In accurate work this is usually defined as

$$\text{GoF}^2[\Phi] = \frac{1}{N_{\text{refl}} - N_{\text{param}}} \sum_{k=1}^{N_{\text{refl}}} \left[\frac{F_k^{\text{calc}}[\Phi] - F_k^{\text{exp}}}{\sigma_k^{\text{exp}}} \right]^2. \quad (20)$$

where:

F_k^{exp} , σ_k^{obs} are the observed X-ray structure factor data, the reflections, of which there are N_{refl} ; and the corresponding estimated experimental uncertainties associated with each reflection; yes, experiments have errors! Sometimes these are also written as F_k^{obs} and σ_k^{obs} , where “obs” is short for observed. Note that the difference between the experimental structure factor and the corresponding calculated value (see below) is weighted by the inverse of the estimated standard deviation; so the value of each term in the sum above is equal to one or smaller if the difference is within one uncertainty.

$F_k^{\text{calc}}[\Phi]$ are the calculated structure factor magnitudes, which I have indicated depend on the particular model wavefunction Φ used—including such things as the basis set used to represent the orbitals, and the type of quantum mechanical calculation used e.g. Hartree-Fock or, these days, some kind of density functional theory (DFT) method. The red Φ indicates that this is the most complicated and expensive term to calculate.

N_{param} is the number of *ad hoc* non-wavefunction parameters which are obtained by least-squares minimisation of GoF^2 , such as the atomic positions, and atomic displacement parameters which are normally produced in any crystallographic structure refinement. N_{param} is subtracted from N_{refl} in the above equation since one data point is “used up” for every parameter determined by the fit procedure; if $N_{\text{param}} = N_{\text{refl}}$ fitting would not make sense since the *ad hoc* parameters exactly fit the data by themselves.

As with any experiment, the more data one has, the better the wavefunction procedure

works. The X-ray diffraction experiment therefore seems ideal because it generates such a large amount of data, and also because there is a famous theorem, *the Hohenberg-Kohn theorem*, which says that the procedure is theoretically well defined (footnote)¹⁴.

Most of the effort in obtaining experimental wavefunctions goes into developing a formula for the calculated structure factors in terms of the wavefunction parameters e.g. the positions of the nuclei $\mathbf{R}$, the molecular orbital coefficients $\mathbf{c}$ for single determinant wavefunctions, or something even more complicated for other wavefunctions.

For completeness, I give the formula for these model structure factors in the cartesian laboratory reference frame.

$$F_k^{\text{calc}}[\Phi] = s |F^{\text{model}}(\mathbf{q}_k)|$$

$$F^{\text{model}}(\mathbf{q}_k) = \sum_A^{\text{atoms}} n_A^{-1} \sum_{\{\mathbf{S}, \mathbf{t}\}}^{\text{symops}} \bar{f}_A(\mathbf{S}^T \mathbf{q}_k) e^{i\mathbf{q}_k \cdot \mathbf{R}_A} e^{i\mathbf{q}_k \cdot \mathbf{t}}$$

where:

$$\begin{aligned} s &= \text{Scale factor} \\ \mathbf{q}_k &= k\text{-th scattering vector} = 2\pi \mathbf{A} \mathbf{h} \\ \mathbf{R}_A &= \text{Asymmetric atom coordinate} \\ \{\mathbf{S}, \mathbf{t}\} &= \text{Roto-translational symmetry operators} \\ n_A &= \text{Atom site symmetry factor} \\ \bar{f}_A(\mathbf{q}_k) &= \text{Thermally-averaged aspherical form factor} \\ &= f_A(\mathbf{q}_k; \{\mathbf{R}_A\}; \Phi) \times \text{TF}(\mathbf{q}_k) \\ f_A(\mathbf{q}_k; \{\mathbf{R}_A\}; \Phi) &= \text{Static aspherical form factor; from } \Psi \\ \text{TF}(\mathbf{k}) &= \text{Temperature factor (Gram-Charlier form)} \\ &= e^{-\frac{1}{2!} \mathbf{U}^2 : \mathbf{k}} \left[1 - \frac{i}{3!} \mathbf{U}^3 : \mathbf{k} + \frac{1}{4!} \mathbf{U}^4 :: \mathbf{k} \right] \\ \mathbf{U}^2, \mathbf{U}^3, \mathbf{U}^4 &= \text{Atomic displacement parameters, ADPs} \end{aligned}$$

The *ad hoc* parameters whose description is in blue are those which are refined by least squares before the constrained wavefunction is obtained. This is generally called “Hirshfeld atom refinement”. Also, I have not explained how the aspherical form factors $f_A(\mathbf{q}_k; \{\mathbf{R}_A\}; \Phi)$ have been calculated from the wavefunction Φ . These are usually obtained by Hirshfeld partitioning. It is a very tricky business; refer to Davidson et al.⁷ for more information. Also, there are alternative schemes that might be proposed for this in the future, see e.g.

¹⁴Indeed, obtaining wavefunctions from electron densities in real space is now not unusual in density functional theory (DFT). The only difference, compared to what is being done above, is that the wavefunction fitting is being done to electron density values in reciprocal space, such values being the structure factor magnitudes.

Chodkiewicz & Wozniak.⁴¹ However, none of these change the basic ideas.

At the very end of this chapter, the equations that need to be solved to minimise the goodness of fit are given; they are very similar to the Hartree-Fock equations.

Constrained (restrained) wavefunctions

Although there is usually plenty enough X-ray data to obtain atomic positions and displacement parameters for most crystalline materials by least squared refinement, there is usually not enough data to obtain the parameters which describe the shape of the electronic wavefunction, e.g. the molecular orbital coefficients $\mathbf{c}$, the so-called electronic degrees of freedom

In order to deal with this problem of not having enough data too determine the parameters in our wavefunction model we must use a method other than least-squares.⁷ These are usually called “regularisation methods” and they work by introducing extra information into via a “constraint” or “restraint” (footnote)¹⁵.

The approximate wavefunction Φ is constrained (or restrained) to produce the desired value of the agreement statistic, by minimising J a combination of the quantum mechanical energy function *and* the least-squares error,

$$J[\Phi] = E_{\text{QM}} + \lambda(\text{GoF}^2 - \Delta^2). \quad (21)$$

The minimisation is with respect to the parameters in Φ *and* the weighting parameter λ .

The reason for having to minimise with respect to λ as an additional parameter may be seen by setting the derivative of J with respect to it equal to zero:

$$\frac{\partial J}{\partial \lambda} = \text{GoF}^2 - \Delta^2 = 0 \implies \text{GoF}^2 = \Delta^2.$$

To solve this equation:

- The value of λ is fixed, and J is minimised. λ is gradually increased from $\lambda = 0$, usually in exponentially increasing values. For each value of λ chosen a different value of the least squares error is obtained, after J is minimised.
- The result is that the least squares error becomes a curve, $\text{GoF}^2(\lambda)$, which may be plotted against λ . It is observed that the least squares error decreases, as expected.

¹⁵I originally used the word “constrained wavefunction”, but A. Genoni has pointed out that the correct word to use for this procedure is “restrained wavefunction”⁴² Unfortunately this incorrect terminology has become embedded in the literature. I am sorry about that. Genoni has also started using the J symbol for the Langrangian, which in the original literature goes by the generic symbol L .

- The procedure is stopped at the value of λ which yields the desired final value of the goodness of fit $\Delta = \text{GoF}$.

To understand more, consider what happens when using the value $\lambda = 0$: in that case $J = E_{\text{QM}}$, so by minimising J we recover the (unfitted) quantum mechanical energy and wavefunction.

From Fig. 6, as $\lambda > 0$ increases, we gradually start to do a least squares fit; that is, the second term in equation (21) starts to get larger relative to E_{QM} .

As λ increases, the minimisation of J gives a value for the least squares error $\text{GoF}^2(\lambda)$ — You can see that this gradually decreases as a function of λ , meaning that the underlying wavefunction Φ starts to fit the data better. Reconstruction is happening!

However, this comes at a cost: the energy of the wavefunction E_{QM} starts to increase. We cannot have it both ways: either the energy is low, when λ is small, or else the least squares error is low, when λ is large.

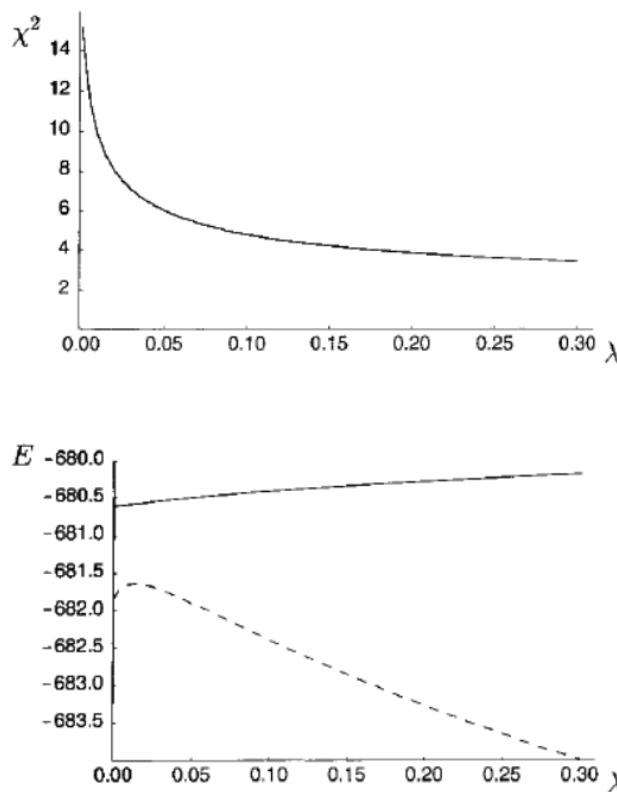


Figure 6: Typical behaviour in an X-ray constrained wavefunction (XCW) model when $J = E_{\text{QM}} + \lambda(\text{GoF}^2 - \Delta)^2$ is minimised to obtain the molecular orbital coefficients in a Hartree-Fock wavefunction Φ . (top) The goodness of fit squared starts to decrease as λ increases. (bottom) The energy E_{QM} starts to increase from its minimum possible value. The dotted line shows the behaviour of the kinetic energy, which is more complicated, but it eventually starts to decrease, indicating that the wavefunction is becoming more kinky.

The halting problem

In principle, with a large basis set, and when λ is large, it seems we should be able to fit the experiment perfectly!

In practice, however, severe numerical problems are encountered. This is because for large values of λ , the problem becomes a least-squares fit, which is ill-conditioned, which is a fancy way of saying that the least-squares problem becomes impossible to solve, as there are too many parameters and not enough data.

Anyway, assuming the uncertainties σ_k^{exp} are reliable, *it does not make sense to make the goodness of fit GoF less than one* since in we would be fitting the data to better than the uncertainty (on average).

Unfortunately, it is known that uncertainties are not reliable. If you think about it, the only way these values could be properly known is if we knew the exact value for the experimental structure factors.

In many cases it is thought that the σ_k^{exp} s are too small, which means to say that the experimental errors are generally larger than estimated. This seems to make sense, since we can't account for all the known sources of experimental error; which is equivalent to saying that the experimental data are more *precise* than *accurate*; see Fig. 7. This means that we should fit the GoF to a value Δ greater than one. However, none of this is really very rigorous .

We are therefore left with a problem.⁷

The halting problem: find the correct method to determine the value of λ , or equivalently the value of $\Delta = \text{GoF}^2(\lambda)$, at which to terminate the constrained fitting procedure.

By “correct method” we mean any logically convincing, practically robust method, which the community accepts.

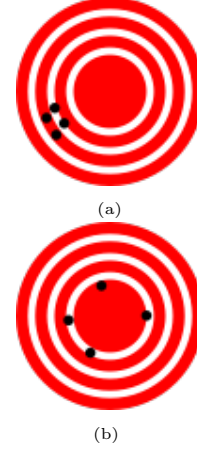


Figure 7: (a) The aim is precise but not very accurate. (b) The aim is accurate but less precise

Possible solutions for the halting problem

There are three viable solutions for the halting problem that have been tried and which work, but none are entirely satisfactory.⁷

1. *The largest possible λ method.*

In this method one chooses the largest possible λ value, or equivalently, the smallest possible value for $\Delta = \text{GoF}$ (as long as the latter value is greater than one) such that numerical convergence problems are not encountered. This is the method most often used in practice. It relies on the idea that the quantum mechanical energy strongly discourages any unphysical wavefunction fitting from occurring, since that energy function is complex and delicate, and therefore easily disrupted by inaccurate or unreasonable experimental data.

Also, the fitted Φ hardly changes even after exponential increases in the value of λ (see below) i.e. for large λ , the fit is not very sensitive. So, this method might be reasonable.

On the other hand, the method seems unsatisfactory because it relies on numerical instabilities, and we do not have a clear understanding of what the source of these instabilities is; we are not sure that they first originate in the quantum mechanical

part; on the other hand, we are sure they are present in the least-squares part, because that is an ill-defined problem.

2. *The exponential fit method.*

It was noticed that the $\text{GoF}^2(\lambda)$ vs. λ curve could always be well fitted by a functional form $Ae^{\lambda B}$; that is, fitted to, a line on a log-log plot. It was found empirically that the $e^{\lambda B}$ curves were very similar over many data sets; and a further theoretical argument showed that at the value of λ where $\Delta = A$ the effective force associated with the quantum mechanical energy is the same as that associated with the least-squares term.

The main criticism of this method is that there was no reason why the form $Ae^{\lambda B}$ should provide a good representation; neither was it an even function, as required by a theoretical analysis.

3. *The Tozer-Ingamells-Handy (TIH) asymptotic method.*

Here the $\text{GoF}^2(\lambda)$ vs. λ curve is fitted to an even asymptotic expansion. From theoretical considerations the constant term in the asymptotic expansion is identified as an inability of the basis-set to fit the data, which was taken as a smoothness criterion on the electron density. Using a three term fit, a formula was derived for the optimum value of λ .

The main criticism concerning this method is that it is rather complex; and it is also not clear whether it is applicable to experimental data, having derived for extrapolating fits to high quality theoretical electron densities.

It was found that the last two methods produced results in remarkable agreement with each other.⁸ However the tests which demonstrated this conclusion, although done on fourteen different data sets, were limited to only one system: α -oxalic acid dihydrate crystal. It is imperative that these tests, or other methods, be reproduced on other molecular crystals before the method is accepted.

Equations for the constrained wavefunction method

I don't want to go into too much detail about the X-ray constrained wavefunctions equations, they are derived elsewhere.⁷ But I do want to say a little about them.

First, if there is no constraint, as expected, we just obtain the usual Hartree-Fock equations for the molecular orbital coefficients $\mathbf{c}$. These take the form of an eigenvalue equation

$$\mathbf{F}\mathbf{c} = \mathbf{S}\mathbf{c}\epsilon. \quad (22)$$

Here $\mathbf{F}$ is an effective Hamiltonian matrix, called the *Fock matrix*, of dimension $N_{\text{bf}} \times N_{\text{bf}}$. The term $\mathbf{F}\mathbf{c}$ is the equation derivative of $E_s\text{QM}$ with respect to $\mathbf{c}$; $\mathbf{S}$ is the overlap

matrix between the basis functions, and the term $S\mathbf{c}\mathbf{c}$ is related to the derivative of the orthonormality condition eqn. (11), which is applied as a constraint (restraint?); and $\mathbf{e}$ are the *molecular orbital energies*. Since $\mathbf{F}$ depends on the MO coefficients, these equations must be solved iteratively, starting from an initial guess, in the way Hartree originally suggested. Standard methods are used, and after a century, are these days highly developed.

When $\lambda > 0$ the only difference is that the Fock matrix changes to include

$$\mathbf{F} \rightarrow \tilde{\mathbf{F}} + \lambda \mathbf{v} \quad (23)$$

and the Hartree-Fock equations remain the same, and are solved in the same way, which makes the procedure very convenient. The expressions for $\mathbf{v}$ are a little complicated, and some tricks are needed to evaluate it efficiently. More important understanding is that the term $\lambda \mathbf{v}\mathbf{c}$ is the derivative of GoF^2 with respect to $\mathbf{c}$, and it is the gradient of the least-squares contribution to J , as might have been guessed by analogy with the term $\mathbf{F}\mathbf{c}$.

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